GID-Power Emission Database (GPED) v1.0

Dataset Overview, Attribute Description & Machine Learning Literature Review

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1. Dataset Overview

The **GID-Power Emission Database (GPED) v1.0** is an open-access, plant-level global emissions database developed by the GID Model research group. Released publicly in **January 2018**, it provides comprehensive information on fossil-fuel- and biomass-burning power-generating units worldwide, including CO_2, SO_2, NO_x, and primary $PM_{2.5}$ emissions data for the reference year 2010.

The database is organized at multiple spatial scales — from individual power plant units to regional and country-level aggregates — making it uniquely suited for global-scale machine learning studies on power sector emissions. An updated version (GPED v1.1) was released in February 2021, covering 2019 emission data.

Official Citation
Zhang, Q. et al. (2018). GID-Power Emission Database (GPED) v1.0. GID Model Research Group. Available at: https://gidmodel.org.cn

2. Dataset Attributes

The following table describes all attributes in GPED v1.0, including an example record from Afghanistan (Jarqodoq power plant) confirming dataset identity.

Attribute	Type	Description	Example Value
No.	Integer	Sequential record identifier	1
Country (or Region)	Categorical	Country or region where the power plant is located	AFGHANISTAN
Plant Name	String	Official name of the power plant	JARQODOQ
Number of Units	Integer	Total generating units at the plant	6
Total Plant Installed Capacity (MW)	Numeric	Aggregate installed capacity in megawatts	15
Fuel Types	Categorical	Primary fuel used (e.g. NG=Natural Gas, COAL, OIL)	NG
CO■ Emissions (Mg)	Numeric	Carbon dioxide emissions in megagrams (tonnes)	74,478.2

SO■ Emissions (Mg)	Numeric	Sulfur dioxide emissions in megagrams	0.3
NOx Emissions (Mg)	Numeric	Nitrogen oxide emissions in megagrams	141.1
PM2.5 Emissions (Mg)	Numeric	Fine particulate matter emissions in megagrams	1.0

Table 1. GPED v1.0 attribute descriptions with example values from the Jarqodoq plant (Afghanistan, Natural Gas, 15 MW).

3. Fuel Type Codes

Code	Fuel Type	Code	Fuel Type
NG	Natural Gas	**COAL**	Coal
OIL	Oil/Petroleum	**BIO**	Biomass
LIGNITE	Lignite	**WASTE**	Waste / Other

4. Research Gap

A systematic search of published literature revealed that **no existing machine learning study has used GPED v1.0 as its primary dataset** for CO_2 emission prediction. The gap is multi-dimensional:

- **Global plant-level scope:** All existing ML studies use national/regional aggregates or single-plant operational data — not global plant-level inventories.
- **Multi-pollutant features:** No published model jointly uses CO_2, SO_2, NO_x, and $PM_{2.5}$ as combined features for CO_2 prediction.
- **Cross-sectional vs. time-series:** Most literature addresses time-series forecasting. GPED v1.0 is cross-sectional (one record per plant) — a different problem formulation requiring classical ML classifiers and regressors.
- **No fold-mix reinforcement strategy** has been applied in any emissions prediction study to date.

> *"To the best of our knowledge, this is the first machine learning study applied to the GID-Power Emission Database v1.0, comprising plant-level CO_2, SO_2, NO_x, and $PM_{2.5}$ emissions across global power generating units."*

5. ML Literature — CO■ Emissions Prediction

The following table summarises the state of the art in machine learning applied to CO_2 and power plant emissions prediction, covering published work from 2020 to 2026. These studies collectively motivate the present work while confirming the absence of GPED v1.0-based ML models.

Reference	Year	Models	Context / Dataset	Best Model	Metrics
Zhu et al.	2024	RF, Deep Forest, XGBoost, Optuna	Coal-fired plant 320 MW, China	**Deep Forest + Optuna**	R^2, MSE, MAE
Multi-Stage Feature Selection Framework (Energies, 2026)	2026	XGBoost, RF, LSTM, SVR	Power station operational data	**XGBoost + SHAP**	R^2, RMSE
Gu et al. (RSC Advances, 2023)	2023	XGBoost, Neural Network, Linear Regression	CEMS + EPA data, New York State	**XGBoost, NN**	R^2, RMSE
Ajala et al. (Environ. Sci. Pollut., 2025)	2025	ARMA, ARIMA, SARMA, SARIMA, SVM, RF, GB, ANN, GRU, LSTM, BiLSTM, CNN-RNN	Daily CO■ — China, India, USA, EU27	**CNN-RNN hybrid**	R^2, MAE, RMSE, MAPE
Carbon metering in power plants (Sci. Reports, 2025)	2025	Linear Regression, XGBoost, LSTM	Coal-fired plant, Haikou China (2024)	**XGBoost**	R^2, RMSE
Energy-based CO■ USA (Sustainability, 2025)	2025	Decision Tree, RF, Linear Regression, KNN, Gradient Boosting, SVR	US EIA energy data 1973–2022	**Gradient Boosting**	R^2, MAE, MSE, RMSE
DPRNN + NiOA (Sci. Reports, 2025)	2025	DPRNN + NiOA, PCA + BSS preprocessing	National CO■ data	**DPRNN + NiOA**	RMSE, R^2
Long-run CO■ & Kuznets curve (ScienceDirect, 2024)	2024	ANN, SVR, Gradient Boosting, RF, XGBoost	National emissions historical data	**XGBoost, NN**	R^2, RMSE
Zeinalipour et al. (arXiv, 2025)	2025	Linear Reg., SVM, Decision Tree, XGBoost, MLP, LSTM, GRU, KNN	Gas turbine emissions (CO, NOx)	**XGBoost, LSTM**	R^2, RMSE, MAE
Comparison ML/DL for CO■ (Inst. Engineers India, 2025)	2025	Gradient Boosting, SVR, RF, LSTM, ANN	National CO■ time-series	**Gradient Boosting**	R^2, MAE, RMSE

Table 2. Summary of published ML/DL models applied to CO■ and power plant emissions prediction (2023–2026). RF = Random Forest; SVR = Support Vector Regression; GB = Gradient Boosting; LSTM = Long Short-Term Memory; ANN = Artificial Neural Network.

6. Key Observations from the Literature

Finding	Implication for This Work

All studies use national/regional or single-plant data	No study uses a global plant-level database like GPED v1.0 — confirmed gap
XGBoost is the most consistently top-performing model	Strong baseline to include; expected to perform well on GPED features
LSTM and CNN-RNN dominate time-series tasks	GPED v1.0 is cross-sectional — classical ML regressors are appropriate
No paper uses SO■, NOx, PM2.5 jointly with CO■ as features	Multi-pollutant feature space is a novel contribution
No paper applies fold-mix reinforcement strategy	Direct methodological novelty connecting to existing Passo Learning work
Metrics: R^2, MAE, RMSE most common	Adopt these as primary metrics + add Sensitivity/Specificity for classification

7. Suggested ML Target Variables

Target Variable	Problem Type	Clinical / Policy Relevance
CO■ Emissions (Mg)	Regression	Most impactful for climate policy and carbon pricing
Fuel Type	Multi-class Classification	Predict fuel type from capacity + emissions pattern
High vs. Low Emitter	Binary Classification	Identify worst offenders for policy targeting
PM2.5 Emissions (Mg)	Regression	Direct public health impact from fine particulates
SO■ Emissions (Mg)	Regression	Acid rain and air quality policy driver

8. References

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